

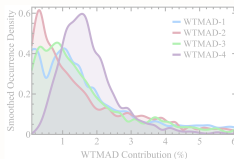
CAMBRIDGE 2026 SEMINAR

DOING MORE WITH LESS:
MINIMALLY EMPIRICAL DFT WITH XDM(Z)



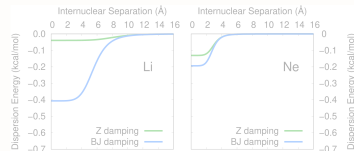
Kyle R. Bryenton
Post-Doctoral Research Associate
Dalhousie Department of Chemistry
September 23, 2026

WTMAD-4: A Fair Weighting Scheme for GMTKN55



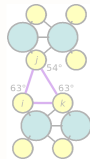
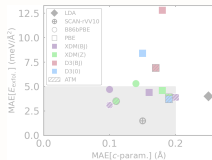
K. R. Bryenton, & E. R. Johnson, *Phys. Chem. Chem. Phys.* **28** (2): 1463–1469. (2026) doi: 10.1039/d5cp03741g

XDM with Z Damping For Molecular Systems



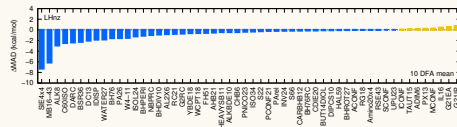
K. R. Bryenton, & E. R. Johnson, *Phys. Chem. Chem. Phys.* **28** (19): 11626–11638. (2026) doi: 10.1039/d6cp00403b

Three-Body XDM(Z) For Layered Materials



A. F. Rumson, K. R. Bryenton, & E. R. Johnson, *Phys. Chem. Chem. Phys.* **28** (22): 13543–13552. (2026) doi: 10.1039/d6cp01304j

A Minimally Empirical Local Hybrid with XDM(Z)



E. R. Johnson & K. R. Bryenton, Submitted to: *J. Chem. Phys.* (2026). doi: 10.48550/arXiv.2609.06162

Introduction

What is Dispersion?

XDM: The Exchange-Hole Dipole Moment Model

Dispersion Damping Functions

The GMTKN55 Database

Open-Source Implementation of XDM(Z)

What's the deal with WTMAD-2?! & XDM(Z) For Molecules

Three-Body XDM(Z) for Layered Materials

Minimally Empirical Local Hybrids

Conclusions

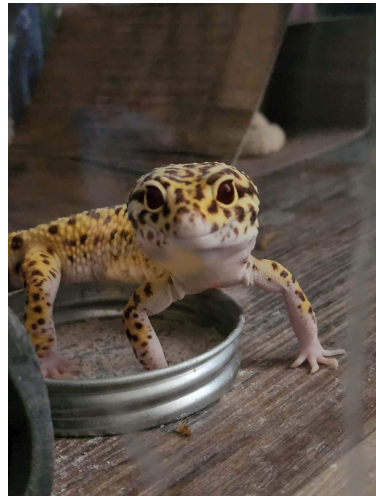
WHAT IS DISPERSION?

London dispersion is a weak, attractive, intermolecular force caused by instantaneous dipole moments between neighbouring atoms.

Properties Determined by Dispersion:

- Adhesion
- Friction
- Surface adsorption
- Phase-change conditions
- Packing in molecular crystals
- Exfoliation of layered materials

Dispersion is absent from nearly every density-functional approximation, so it must be added back in.



Scamper the Gecko

THE XDM MODEL

The exchange-hole dipole moment (XDM) model calculates the dispersion energy via a pairwise sum over all atoms,

$$E_{\text{XDM}} = - \sum_{i < j} \sum_{n=6,8,10} f_n^{\text{damp}}(R_{ij}) \frac{C_{n,ij}}{R_{ij}^n}.$$

Here, $f_n^{\text{damp}}(R_{ij})$ is a damping function to prevent divergence at small interatomic separations, R_{ij} .

To obtain the dispersion coefficients, C_n , first we compute the exchange-hole dipole moment,

$$\mathbf{d}_{\text{X}\sigma}(\mathbf{r}) = \left[\int \mathbf{s} \left| h_{\text{X}\sigma}(\mathbf{r}, \mathbf{s}) \right| d\mathbf{s} \right] - \mathbf{r},$$

which can be obtained by integrating the exchange hole, $h_{\text{X}\sigma}$, over its reference point, $\mathbf{s}$. The dispersion coefficients may be calculated from these moments and the atom-in-molecule polarisabilities α_i . e.g.,

$$C_{6,ij}^{\text{XDM}} = \frac{\alpha_i \alpha_j \langle d_{\text{X}}^2 \rangle_i \langle d_{\text{X}}^2 \rangle_j}{\alpha_i \langle d_{\text{X}}^2 \rangle_j + \alpha_j \langle d_{\text{X}}^2 \rangle_i}$$

BJ DAMPING

Canonically, XDM has damped the dispersion energy to a finite, non-zero value via use of the Becke–Johnson damping function,

$$f_n^{\text{BJ}}(R_{ij}) = \frac{R_{ij}^n}{R_{ij}^n + (a_1 R_{c,ij} + a_2)^n}.$$

$R_{c,ij}$ is a “critical” interatomic distance at which successive terms in the perturbation theory expansion of the dispersion energy become equal:

$$R_{c,ij} = \frac{1}{3} \left[\sqrt{\frac{C_{8,ij}}{C_{6,ij}}} + \sqrt[4]{\frac{C_{10,ij}}{C_{6,ij}}} + \sqrt{\frac{C_{10,ij}}{C_{8,ij}}} \right].$$

The damping parameters, a_1 and a_2 , are optimised by minimising the RMS percent error in binding energies for a set of 49 molecular dimers, KB49.

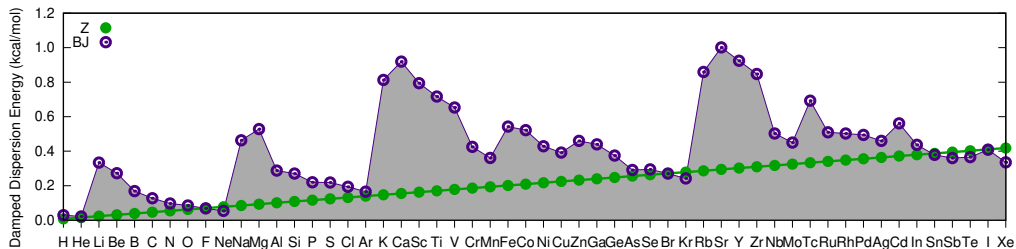
Johnson & Becke, *J. Chem. Phys.* **124**, 174104. (2006) doi: 10.1063/1.2190220

Kannemann & Becke, *J. Chem. Theory Comput.* **6**, 1081–1088 (2010) doi: 10.1021/ct900699r

BECKE'S ONE-PARAMETER DAMPING FUNCTION: Z DAMPING

To resolve ALK8 issues seen with BJ damping when working on his DH24 double-hybrid functional, Becke proposed a new damping function with only one empirical parameter. This new form has the advantage of recovering correlation energies roughly proportional to atomic number in the united-atom limit.

$$f_n^Z(R_{ij}) = \frac{R_{ij}^n}{R_{ij}^n + z_{\text{damp}} \frac{C_{n,ij}}{Z_i + Z_j}} \quad \lim_{R_{ij} \rightarrow 0} E_{\text{XDM}(Z)} = \frac{3}{z_{\text{damp}}} (Z_i + Z_j)$$



Becke, *J. Chem. Phys.* **160**, 204118. (2024) doi: 10.1063/5.0207682

Bryenton & Johnson, *Phys. Chem. Chem. Phys.* **28**, 11626–11638. (2026) doi: 10.1039/D6CP00403B

Rumson, Bryenton, & Johnson, *Phys. Chem. Chem. Phys.* **28**, 13543–13552. (2026) doi: 10.1039/D6CP01304J

THE GMTKN55 DATABASE

Database for general main group thermochemistry, kinetics, and noncovalent interactions

GMTKN55				
AL2X6	NBPRC	ISO34	WCPT18	S66
ALK8	PA26	ISOL24	ADIM6	WATER27
ALKBDE10	RC21	MB16-43	AHB21	ACONF
BH76RC	SIE4x4	PArel	CARBHB12	Amino20x4
DC13	TAUT15	RSE43	CHB6	BUT14DIOL
DIPCS10	W4-11	BH76	HAL59	ICONF
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HEAVYSB11	DARC	PX13	S22	UPU23

Five major categories: basic properties and reaction energies for small systems, reaction energies for large systems and isomerisation reactions, barrier heights, intermolecular noncovalent interactions, and intramolecular noncovalent interactions

$$WTMAD-2 = \frac{1}{N_{\text{bench}}} \sum_{i=1}^{N_{\text{bench}}} \underbrace{\frac{N_{\text{syst},i}}{N_{\text{syst,mean}}}}_{\text{\# of Systems}} \cdot \underbrace{\frac{\overline{|\Delta E|}_{\text{mean}}}{|\Delta E|_i}}_{\text{Mean Ref. En.}} \cdot MAD_i$$

$w_i^{\text{WTMAD-2}}$

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of Systems
Mean Ref. En.

OPEN-SOURCE IMPLEMENTATION AND TESTING OF XDM(Z)

PCCP



PAPER



Cite this: DOI: 10.1039/d6cp00403b

Consistent GMTKN55 and molecular-crystal accuracy using minimally empirical DFT with XDM(Z) dispersion

Kyle R. Bryenton ^{ab} and Erin R. Johnson ^{*abc}

Bryenton & Johnson, *Phys. Chem. Chem. Phys.* **28**, 11626–11638. (2026) doi: 10.1039/D6CP00403B

Bryenton & Johnson, *Phys. Chem. Chem. Phys.* **28**, 1463–1469. (2026) doi: 10.1039/d5cp03741g

~~OPEN-SOURCE IMPLEMENTATION AND TESTING OF XDM(Z)~~
SIDE QUEST: FIGURE OUT THE WEIRD THING HAPPENING WITH WTMAD-2





ROYAL SOCIETY OF CHEMISTRY

PAPER

WTMAD-4: a fair weighting scheme for GMTKN55

Cite this: *Phys. Chem. Chem. Phys.*, 2026, **28**, 1463

Kyle R. Bryenton ^{ab} and Erin R. Johnson ^{*abc}

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Introduction

What's the deal with WTMAD-2?! & XDM(Z) For Molecules

- The issue with WTMAD-2

- Definition of WTMAD-4

- Improvements in Statistics seen with WTMAD-4

- Where does Z Damping Help?

- Outlier Analysis

- TS and MBD on GMTKN55

- Transferability to Molecular Crystals

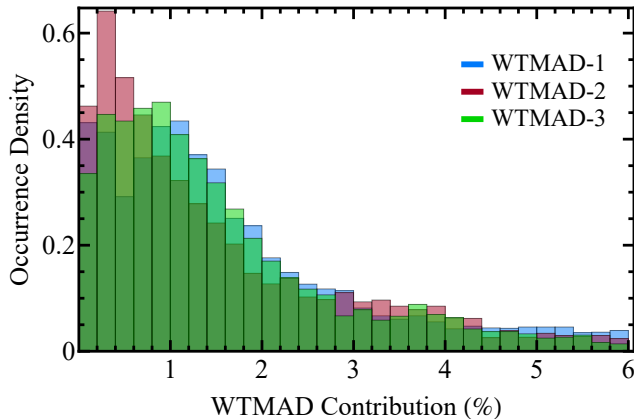
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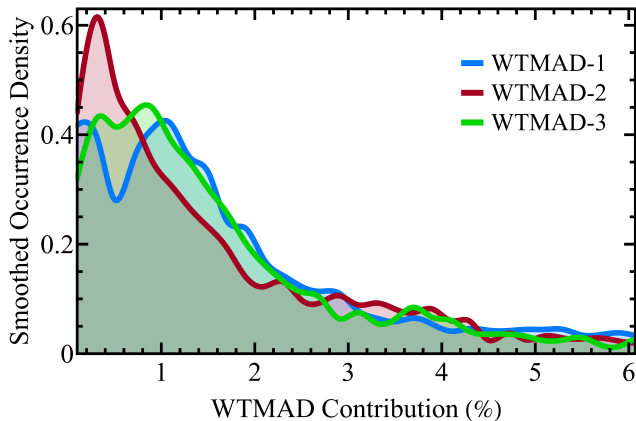
WTMAD-2 WOES

- Statistics don't form a Gaussian distribution, and have a long decaying tail
- The top 3 benchmarks contribute as much to WTMAD-2 as the bottom 36
- Benchmarks are weighed, on average, $200\times$ greater than others
- Consistent underweighting of some benchmarks (IL16: 0.06%) and overweighting of others (BH76: 9.89%)
- Poses issues for ML or parameter optimisation based on WTMAD-2

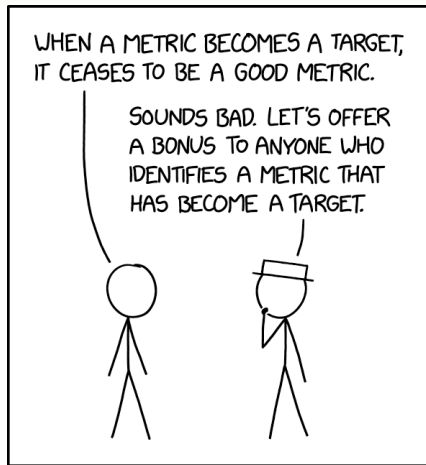


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GOODHART'S LAW



*[later] I'm pleased to report we're now identifying
and replacing hundreds of outdated metrics per hour.*

WTMAD-2 vs WTMAD-4

- These issues are caused by WTMAD-2's dependence on the mean reaction reference energies
- Some benchmarks have $|\overline{\Delta E}|_i \approx 100$ kcal/mol with *small* average errors
- Other benchmarks have $|\overline{\Delta E}|_i \approx 1$ kcal/mol with *huge* average errors

$$\text{WTMAD-}N = \frac{1}{N_{\text{bench}}} \sum_{i=1}^{N_{\text{bench}}} w_i^{\text{WTMAD-}N} \cdot \text{MAD}_i$$

$$w_i^{\text{WTMAD-2}} = \frac{N_{\text{syst},i}}{N_{\text{syst,mean}}} \frac{|\overline{\Delta E}|_{\text{mean}}}{|\overline{\Delta E}|_i}$$

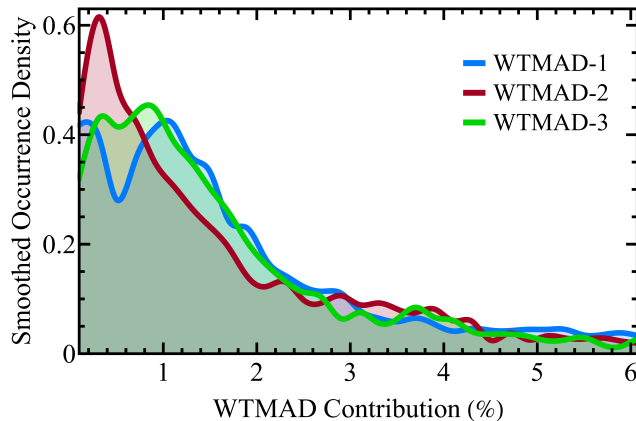
$$w_i^{\text{WTMAD-4}} = \frac{100}{N_{\text{bench}}} \frac{3.5}{\text{MAD}_i^{10\text{-DFA}}}$$

The “10-DFA” List (incl. DFT-D3 Dispersion):

PBE0	B3LYP	B3P86	PW1PW91	HSE06
HISS	BHLYP	B3PW91	MPW1PW91	LC- ω pPBE

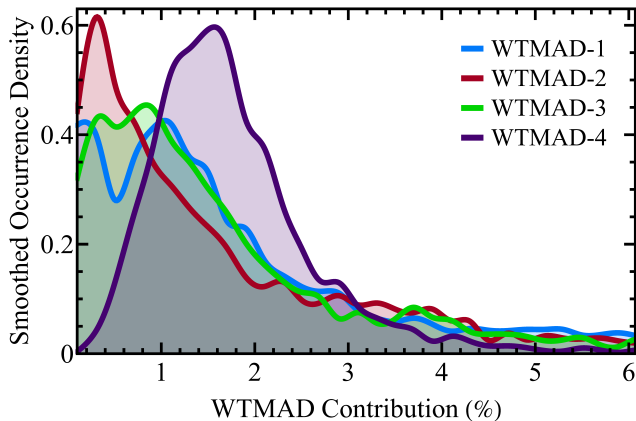
WTMAD-4 Wows

- Statistics form a more Gaussian-like distribution centred at $100\%/55 = 1.8\%$
- The top 3 benchmarks contribute as much to WTMAD-4 as the bottom 6
- Benchmarks are weighed typically within a factor of 3
- WTMAD-4 is consistent across all rungs
- Fairer treatment of benchmarks for ML uses or parameter optimisation



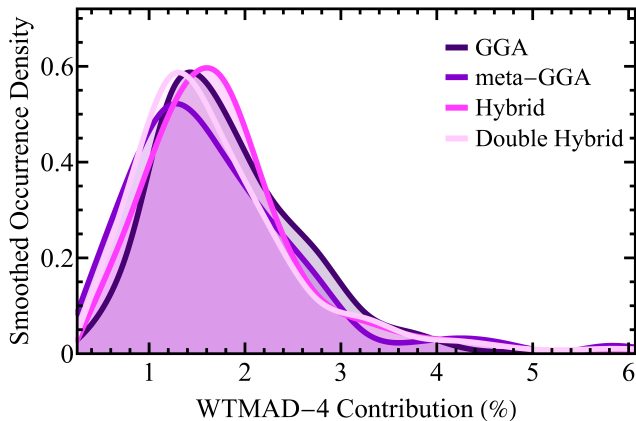
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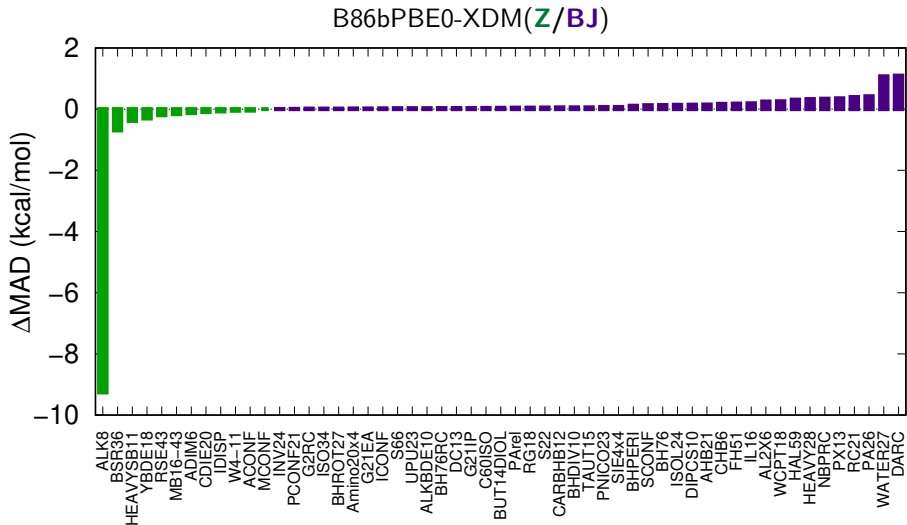


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RETURNING TO Z DAMPING: WHERE DOES IT HELP?



GGAs AND META-GGA FUNCTIONALS

DEFINITIONS:

$N_{r>2}$: Number of benchmarks with an error $> 2\times$ *ratio* compared to the “10-DFA” mean

$N_{d>2}$: Number of benchmarks with an error > 2 kcal/mol *difference* compared to the “10-DFA” mean

Functional	WTMAD- N			$N_{r>2}$	MAX r	Set	$N_{d>2}$	MAX d	Set
	2	4	$N_{r\leq 1}$						
revPBE-D3(BJ)	8.24	7.67	16	2	2.22	BHPERI	9	9.70	SIE4x4
revPBE-XDM(Z)	8.69	7.91	19	7	2.51	SCONF	8	9.65	SIE4x4
revPBE-V	8.50	7.94	23	8	2.88	DIPCS10	11	9.36	MB16-43
OLYP-D3(BJ)	8.76	8.16	18	6	2.51	BHPERI	13	11.95	SIE4x4
B97-D3(BJ)	8.49	8.32	19	6	2.57	AL2X6	12	16.80	MB16-43
revPBE-XDM(BJ)	8.40	8.45	16	7	3.76	ACONF	11	9.79	SIE4x4
B97M-V	5.46	5.24	39	0	1.86	MB16-43	2	16.67	MB16-43
B97M-D4	5.68	5.44	37	0	1.95	MB16-43	2	18.24	MB16-43
B97M-D3(BJ)	6.44	6.41	33	3	2.81	AL2X6	3	4.97	ALK8
r^2 SCAN-D3(BJ)	7.12	6.79	25	0	1.72	WCPT18	6	4.57	SIE4x4
B97M-D3(0)	6.51	6.92	31	3	4.39	ADIM6	5	28.23	MB16-43
revTPSS-D3(BJ)	8.44	7.73	23	3	3.71	SCONF	9	17.45	MB16-43

GGAs: revPBE is still the best GGA functional for molecular chemistry
 mGGAs: Most meta-GGAs performed worse than GGAs, and often with much larger max errors

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revPBE-V	8.50	7.94	23	8	2.88	DIPCS10	11	9.36	MB16-43
OLYP-D3(BJ)	8.76	8.16	18	6	2.51	BHPERI	13	11.95	SIE4x4
B97-D3(BJ)	8.49	8.32	19	6	2.57	AL2X6	12	16.80	MB16-43
revPBE-XDM(BJ)	8.40	8.45	16	7	3.76	ACONF	11	9.79	SIE4x4
B97M-V	5.46	5.24	39	0	1.86	MB16-43	2	16.67	MB16-43
B97M-D4	5.68	5.44	37	0	1.95	MB16-43	2	18.24	MB16-43
B97M-D3(BJ)	6.44	6.41	33	3	2.81	AL2X6	3	4.97	ALK8
r^2 SCAN-D3(BJ)	7.12	6.79	25	0	1.72	WCPT18	6	4.57	SIE4x4
B97M-D3(0)	6.51	6.92	31	3	4.39	ADIM6	5	28.23	MB16-43
revTPSS-D3(BJ)	8.44	7.73	23	3	3.71	SCONF	9	17.45	MB16-43

GGAs: revPBE is still the best GGA functional for molecular chemistry

mGGAs: Most meta-GGAs performed worse than GGAs, and often with much larger max errors

HYBRID AND RANGE-SEPARATED HYBRID FUNCTIONALS

Functional	WTMAD- N			$N_{r>2}$	MAX r	Set	$N_{d>2}$	MAX d	Set
	2	4	$N_{r\leq 1}$						
PW6B95-D3(BJ)	5.50	5.14	41	1	2.31	IL16	0	1.76	SIE4x4
M05-2X-D3(0)	4.62	5.19	41	1	2.08	ADIM6	3	7.02	MB16-43
M06-2X-D3(0)	4.92	5.22	46	2	3.50	HEAVYSB11	1	5.83	HEAVYSB11
PW6B95-V	5.57	5.24	45	2	2.49	ADIM6	0	1.81	SIE4x4
revPBE0-D3(BJ)	5.31	5.30	43	0	1.51	W4-11	1	3.03	W4-11
revPBE0-XDM(Z)	5.71	5.43	40	0	1.48	W4-11	1	2.83	W4-11
revPBE0-XDM(BJ)	5.06	5.44	42	1	2.58	ALK8	2	7.71	ALK8
MPW1B95-D3(BJ)	5.55	5.48	42	1	2.02	ACONF	0	0.85	RSE43
B86bPBE0-XDM(BJ)	5.77	5.53	39	0	1.93	ALK8	1	4.55	ALK8
M08-HX-D3(0)	5.27	5.68	41	2	3.85	ACONF	1	2.26	C60ISO
MPW1PW91-D3(BJ)	6.33	5.82	32	0	1.31	PNICO23	0	1.04	ALK8
B86bPBE0-XDM(Z)	6.43	5.85	39	0	1.55	SCONF	0	1.68	WATER27
ω B97M-V	3.18	3.72	49	1	2.22	C60ISO	1	6.51	C60ISO
ω B97M-D4	4.02	4.34	43	1	2.24	C60ISO	2	6.60	C60ISO
ω B97X-V	3.93	4.40	50	1	2.57	C60ISO	2	13.21	MB16-43
ω B97M-D3(BJ)	3.94	4.60	44	3	2.47	C60ISO	2	7.86	C60ISO
ω B97X-D4	4.33	4.68	46	1	2.65	C60ISO	1	8.83	C60ISO

Hybrids: Significant reordering caused by the worsening of mGGA-based hybrids and the improvement of many minimally-empirical ones

RS Hybrids: The ω B97 family performs excellently, but error increased due to large MADs on C60ISO and MB16-43 benchmarks, which WTMAD-2 underweighted

HYBRID AND RANGE-SEPARATED HYBRID FUNCTIONALS

Functional	WTMAD- N			$N_{r>2}$	MAX r	Set	$N_{d>2}$	MAX d	Set
	2	4	$N_{r\leq 1}$						
PW6B95-D3(BJ)	5.50	5.14	41	1	2.31	IL16	0	1.76	SIE4x4
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PW6B95-V	5.57	5.24	45	2	2.49	ADIM6	0	1.81	SIE4x4
revPBE0-D3(BJ)	5.31	5.30	43	0	1.51	W4-11	1	3.03	W4-11
revPBE0-XDM(Z)	5.71	5.43	40	0	1.48	W4-11	1	2.83	W4-11
revPBE0-XDM(BJ)	5.06	5.44	42	1	2.58	ALK8	2	7.71	ALK8
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ω B97M-D4	4.02	4.34	43	1	2.24	C60ISO	2	6.60	C60ISO
ω B97X-V	3.93	4.40	50	1	2.57	C60ISO	2	13.21	MB16-43
ω B97M-D3(BJ)	3.94	4.60	44	3	2.47	C60ISO	2	7.86	C60ISO
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	2	4	$N_{r\leq 1}$						
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ω B97X-V	3.93	4.40	50	1	2.57	C60ISO	2	13.21	MB16-43
ω B97M-D3(BJ)	3.94	4.60	44	3	2.47	C60ISO	2	7.86	C60ISO
ω B97X-D4	4.33	4.68	46	1	2.65	C60ISO	1	8.83	C60ISO

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DOUBLE HYBRIDS AND MACHINE-LEARNED FUNCTIONALS

Functional	WTMAD- N			$N_{r>2}$	MAX r	Set	$N_{d>2}$		Set
	2	4	$N_{r\leq 1}$				MAX d		
DH24	1.72	2.23	52	0	1.57	C60ISO	1	3.07	C60ISO
revDH23	1.72	2.26	52	0	1.56	C60ISO	1	2.99	C60ISO
SOS-DH24	1.91	2.27	52	0	1.26	IL16	0	0.66	C60ISO
SOS-DH23	1.95	2.34	52	0	1.35	IL16	0	0.73	C60ISO
ω DOD-PBEP86-D3(BJ)	2.20	2.50	53	0	1.35	ADIM6	0	0.73	DIPCS10
ω DOD-PBEP86-D3(BJ)	2.21	2.52	53	0	1.54	ADIM6	0	0.30	DIPCS10
DM21mu	3.93	3.72	50	1	2.91	RG18	0	1.07	HEAVYSB11
DM21	3.97	3.94	48	2	2.68	RG18	1	6.01	C60ISO
DM21mc	3.96	3.98	49	1	2.12	ACONF	0	1.25	C60ISO
Skala	3.83	4.01	47	0	1.60	C60ISO	1	3.20	C60ISO
DM21m	3.89	4.04	48	1	2.98	ACONE	0	0.82	HEAVYSB11

D Hybrids: DH24 keeps the top spot, while XYG8 substantially drops from 3rd to 17th due to WTMAD-2 parametrisation

ML DFAs: Perform on par with the best range-separated hybrids, while seemingly having less extreme outliers compared to other highly-parametrised functionals of a similar rung

DOUBLE HYBRIDS AND MACHINE-LEARNED FUNCTIONALS

Functional	WTMAD- N			$N_{r>2}$	MAX r	Set	$N_{d>2}$		Set
	2	4	$N_{r\leq 1}$				MAX d		
DH24	1.72	2.23	52	0	1.57	C60ISO	1	3.07	C60ISO
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SOS-DH24	1.91	2.27	52	0	1.26	IL16	0	0.66	C60ISO
SOS-DH23	1.95	2.34	52	0	1.35	IL16	0	0.73	C60ISO
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SOS-DH24	1.91	2.27	52	0	1.26	IL16	0	0.66	C60ISO
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ω DOD-PBEP86-D3(BJ)	2.21	2.52	53	0	1.54	ADIM6	0	0.30	DIPCS10
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TS, MBD@rsSCS, AND MBD-NL ON GMTKN55

This work is also the first time the TS and MBD corrections have been tested on GMTKN55. They were paired with PBE and PBE0 only, due to the limited availability of damping parameters.

WTMAD-4	TS	MBD@rsSCS	MBD-NL	XDM(BJ)	XDM(Z)
PBE	9.85	9.44*	9.42*	9.20	9.27
PBE0	6.72	6.30*	6.36*	6.21	6.19

* Two systems failed to converge for each of MBD@rsSCS and MBD-NL as a result of polarisation catastrophe issues.

MBD INSIGHTS:

- TS shows its largest errors for isomerisations and large systems
- For ALK8, MBD@rsSCS shows similar errors to XDM(BJ)
- PBE0-MBD-NL generally has good performance with no outliers and a MAE of 14.86 on MB16-43
- However, we don't recommend these MBD methods for general thermochemistry

TRANSFERABILITY TO MOLECULAR CRYSTALS?

MOLECULAR CRYSTAL BENCHMARKS

X23 : 23 molecular crystals

HalCrys4 : 4 halogen-bonded crystals

ICE13 : 13 ice polymorphs (absolute and relative)

Basis-set corrected MAEs, in kcal/mol

Functional	X23		HalCrys4		ICE13-Abs		ICE13-Rel	
	BJ	Z	BJ	Z	BJ	Z	BJ	Z
PBE	1.31	0.92	5.49	4.12	1.44	2.10	0.82	0.61
revPBE	1.27	1.30	3.55	4.69	0.30	0.30	0.39	0.30
B86bPBE	0.70	0.81	4.70	5.03	1.56	1.88	0.52	0.41
PBE0	1.00	0.66	1.61	0.57	0.43	0.50	0.48	0.29
revPBE0	0.90	0.71	2.47	1.12	1.36	0.79	0.23	0.21
B86bPBE0	0.48	0.61	1.21	0.86	0.30	0.36	0.31	0.17
PBE50	0.87	0.75	1.78	3.78	1.30	0.69	0.21	0.24
revPBE50	0.72	0.74	2.81	3.71	2.18	1.41	0.18	0.39
B86bPBE50	0.51	0.73	1.10	3.36	1.25	0.73	0.18	0.33

- GGAs overbind substantially for HalCrys4 due to delocalisation error
- The 25% hybrids seem the mostly consistent across the benchmarks
- But XDM(Z) is good paired with any of PBE0, revPBE0, or B86bPBE0
- Considering the GMTKN55 data, revPBE0-XDM(Z), and B86bPBE0-XDM(Z) seem the most transferable (so far)

Introduction

What's the deal with WTMAD-2?! & XDM(Z) For Molecules

Three-Body XDM(Z) for Layered Materials

- The LM26 Benchmark

- The ATM Three-Body Dispersion Term

- Three-Body Dispersion Damping Functions

- LM26 Results

Minimally Empirical Local Hybrids

Conclusions

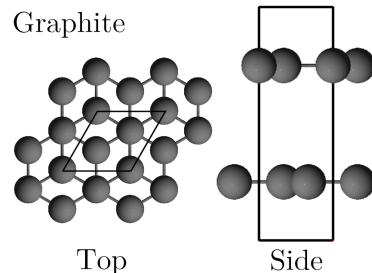
LAYERED MATERIALS AND EXFOLIATION ENERGIES

THE LM26 BENCHMARK

Random-phase approximation (RPA) reference exfoliation energies and c lattice parameters for 26 layered materials including

- Graphite
- Hexagonal Boron Nitride
- Lead(II) Oxide
- Transition-metal dichalcogenides

Note: Recent QMC studies for graphite, MoS_2 and TiS_2 suggest the RPA exfoliation energies are 5–10 meV/Å² too small.



These dispersion-bound materials are important for applications in 2D semiconductors, optoelectronics, sensors, energy storage, and solid-state lubricants.

$$E_{\text{exfol}} = -\frac{1}{A} \left(\frac{E_{\text{bulk}}}{N_{\text{layers}}} - E_{\text{mono}} \right)$$

Björkman, *J. Chem. Phys.* **141**, 074708. (2014) doi: 10.1063/1.4893329

Rumson, **Bryenton**, & Johnson, *Phys. Chem. Chem. Phys.* **28**, 13543–13552. (2026) doi: 10.1039/D6CP01304J

THE ATM THREE-BODY DISPERSION TERM

The leading term in the three-body contribution to dispersion is the Axilrod–Teller–Muto (ATM) energy for an atom triple ijk :

$$E_{\text{ATM}} = \sum_{i < j < k} \frac{C_{9,ijk} f_{\text{ATM}}^{\text{damp}}}{(R_{ij} R_{ik} R_{jk})^3} (3 \cos \theta_i \cos \theta_j \cos \theta_k + 1)$$

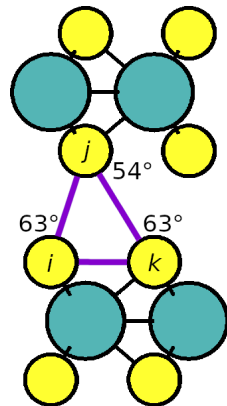
The angular dependence predicts:

Linear: maximally attractive

Right: repulsive

Equilateral: maximally repulsive

For molecular dimers and crystals, ATM is usually negligible. But layered materials have a unique geometry that can generate a large, repulsive three-body interaction between the layers.



A MoS₂ bilayer: near-equilateral triples span adjacent layers.

Axilrod & Teller, *J. Chem. Phys.* **11**, 299–300. (1943) doi: 10.1063/1.1723844

Muto, *Proc. Phys. Math. Soc. Jpn.* **17**, 629–631. (1943) doi: 10.11429/subutsukaishi1927.17.10-11-12.629

THREE-BODY DISPERSION DAMPING FUNCTIONS

For BJ damping, the recommended form is a product of three pairwise BJ functions of order 3. A similar Z damping analogue is proposed:

$$f_{\text{ATM}}^{\text{BJ}} = f_3^{\text{BJ}}(R_{ij}) f_3^{\text{BJ}}(R_{ik}) f_3^{\text{BJ}}(R_{jk})$$

$$f_{\text{ATM}}^{\text{Z}} = f_{\text{partial}}^{\text{Z}}(R_{ij}) f_{\text{partial}}^{\text{Z}}(R_{ik}) f_{\text{partial}}^{\text{Z}}(R_{jk})$$

$$f_3^{\text{BJ}}(R_{ij}) = \frac{R_{ij}^3}{R_{ij}^3 + (a_1 R_{c,ij} + a_2)^3}$$

$$f_{\text{partial}}^{\text{Z}}(R_{ij}) = \frac{R_{ij}^3}{R_{ij}^3 + \sqrt{\frac{z_{\text{damp}} C_{6,ij}}{Z_i + Z_j}}}$$

A square root appears in the denominator to insure the correct united-atom limit. Using the standard approximation $C_{9,ijk} \approx \sqrt{C_{6,ij} C_{6,ik} C_{6,jk}} \cdot \text{Ha}^{-1/2}$, the united-atom limit becomes

$$\lim_{\forall R \rightarrow 0} \frac{f_{\text{ATM}}^{\text{Z}} C_{9,ijk}}{R_{ij}^3 R_{ik}^3 R_{jk}^3} \approx \left(\frac{1}{\sqrt{\text{Ha}}} \right) \frac{\sqrt{(Z_i + Z_j)(Z_i + Z_k)(Z_j + Z_k)}}{z_{\text{damp}}^{3/2}},$$

which is effectively independent of the dispersion coefficients exactly the property that initially motivated Z damping.

LM26: EXFOLIATION ENERGIES

MAE and ME, in meV/Å². Positive ME = overbinding.

Functional	lightdenser		tight		PW	
	MAE	ME	MAE	ME	MAE	ME
revPBE-D3(BJ)	35.1	35.1	35.0	35.0	34.5	34.5
revPBE-D3(BJ)+ATM	27.8	27.8	27.7	27.0	27.2	27.2
revPBE-D3(0)	16.5	16.5	16.2	16.2	15.8	15.8
revPBE-D3(0)+ATM	10.2	10.0	9.9	9.6	9.6	9.2
PBE-D3(BJ)	14.7	14.6	14.5	14.3	12.8	12.6
PBE-D3(BJ)+ATM	8.2	7.8	8.1	7.5	6.9	6.1
PBE-D3(0)	9.8	9.6	9.5	9.2	8.4	8.0
PBE-D3(0)+ATM	4.1	3.2	4.0	2.8	3.7	1.8
revPBE-XDM(BJ)	5.0	5.0	11.5	11.5	6.4	4.7
revPBE-XDM(Z)	6.5	6.1	8.2	7.8	5.4	4.2
revPBE-XDM(BJ)+ATM	2.2	-0.9	5.2	5.2	4.4	-2.2
revPBE-XDM(Z)+ATM	4.6	-0.3	5.3	0.8	5.3	-2.0
B86bPBE-XDM(BJ)	5.9	5.9	7.7	7.7	4.7	3.3
B86bPBE-XDM(Z)	5.8	5.5	7.3	7.1	5.3	4.5
B86bPBE-XDM(BJ)+ATM	2.3	1.2	3.5	3.2	3.1	-0.2
B86bPBE-XDM(Z)+ATM	3.0	1.3	4.0	2.6	3.5	0.4
PBE-XDM(BJ)	5.6	5.5	6.8	6.7	4.4	1.6
PBE-XDM(Z)	4.1	3.2	5.4	4.8	4.6	1.8
PBE-XDM(BJ)+ATM	2.7	0.9	3.3	2.1	3.9	-2.2
PBE-XDM(Z)+ATM	2.6	-0.1	3.2	1.0	3.9	-1.4
revPBE	19.8	-19.8	20.1	-20.1	20.3	-20.3
B86bPBE	18.6	-18.6	19.2	-19.2	19.5	-19.5
PBE	17.7	-17.7	18.3	-18.3	18.6	-18.6
PBEsol	11.9	-11.9	12.1	-12.1	12.7	-12.7
LDA	3.7	-1.9	4.0	-2.2	4.0	-2.4

- Calculated using FHI-aims (lightdenser / tight) and Quantum ESPRESSO (planewave basis)
- Bare GGAs underbind due to lack of dispersion
- LDA gives reasonable energies through fortuitous error cancellation and underestimates the c
- D3 overbinds from overestimated dispersion coefficients
- XDM is more consistently accurate, but still tends to overbind. BJ and Z give the same average MAE: 4.9 meV/Å²
- The large and repulsive ATM term improves both the ME and MAE
- Clearly B86bPBE-XDM(BJ/Z)+ATM and PBE-XDM(BJ/Z)+ATM are the best performing methods

LM26: EXFOLIATION ENERGIES

MAE and ME, in meV/Å². Positive ME = overbinding.

Functional	lightdenser		tight		PW	
	MAE	ME	MAE	ME	MAE	ME
revPBE-D3(BJ)	35.1	35.1	35.0	35.0	34.5	34.5
revPBE-D3(BJ)+ATM	27.8	27.8	27.7	27.0	27.2	27.2
revPBE-D3(0)	16.5	16.5	16.2	16.2	15.8	15.8
revPBE-D3(0)+ATM	10.2	10.0	9.9	9.6	9.6	9.2
PBE-D3(BJ)	14.7	14.6	14.5	14.3	12.8	12.6
PBE-D3(BJ)+ATM	8.2	7.8	8.1	7.5	6.9	6.1
PBE-D3(0)	9.8	9.6	9.5	9.2	8.4	8.0
PBE-D3(0)+ATM	4.1	3.2	4.0	2.8	3.7	1.8
revPBE-XDM(BJ)	5.0	5.0	11.5	11.5	6.4	4.7
revPBE-XDM(Z)	6.5	6.1	8.2	7.8	5.4	4.2
revPBE-XDM(BJ)+ATM	2.2	-0.9	5.2	5.2	4.4	-2.2
revPBE-XDM(Z)+ATM	4.6	-0.3	5.3	0.8	5.3	-2.0
B86bPBE-XDM(BJ)	5.9	5.9	7.7	7.7	4.7	3.3
B86bPBE-XDM(Z)	5.8	5.5	7.3	7.1	5.3	4.5
B86bPBE-XDM(BJ)+ATM	2.3	1.2	3.5	3.2	3.1	-0.2
B86bPBE-XDM(Z)+ATM	3.0	1.3	4.0	2.6	3.5	0.4
PBE-XDM(BJ)	5.6	5.5	6.8	6.7	4.4	1.6
PBE-XDM(Z)	4.1	3.2	5.4	4.8	4.6	1.8
PBE-XDM(BJ)+ATM	2.7	0.9	3.3	2.1	3.9	-2.2
PBE-XDM(Z)+ATM	2.6	-0.1	3.2	1.0	3.9	-1.4
revPBE	19.8	-19.8	20.1	-20.1	20.3	-20.3
B86bPBE	18.6	-18.6	19.2	-19.2	19.5	-19.5
PBE	17.7	-17.7	18.3	-18.3	18.6	-18.6
PBEsol	11.9	-11.9	12.1	-12.1	12.7	-12.7
LDA	3.7	-1.9	4.0	-2.2	4.0	-2.4

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MAE and ME, in meV/Å². Positive ME = overbinding.

Functional	lightdenser		tight		PW	
	MAE	ME	MAE	ME	MAE	ME
revPBE-D3(BJ)	35.1	35.1	35.0	35.0	34.5	34.5
revPBE-D3(BJ)+ATM	27.8	27.8	27.7	27.0	27.2	27.2
revPBE-D3(0)	16.5	16.5	16.2	16.2	15.8	15.8
revPBE-D3(0)+ATM	10.2	10.0	9.9	9.6	9.6	9.2
PBE-D3(BJ)	14.7	14.6	14.5	14.3	12.8	12.6
PBE-D3(BJ)+ATM	8.2	7.8	8.1	7.5	6.9	6.1
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PBE-D3(0)+ATM	4.1	3.2	4.0	2.8	3.7	1.8
revPBE-XDM(BJ)	5.0	5.0	11.5	11.5	6.4	4.7
revPBE-XDM(Z)	6.5	6.1	8.2	7.8	5.4	4.2
revPBE-XDM(BJ)+ATM	2.2	-0.9	5.2	5.2	4.4	-2.2
revPBE-XDM(Z)+ATM	4.6	-0.3	5.3	0.8	5.3	-2.0
B86bPBE-XDM(BJ)	5.9	5.9	7.7	7.7	4.7	3.3
B86bPBE-XDM(Z)	5.8	5.5	7.3	7.1	5.3	4.5
B86bPBE-XDM(BJ)+ATM	2.3	1.2	3.5	3.2	3.1	-0.2
B86bPBE-XDM(Z)+ATM	3.0	1.3	4.0	2.6	3.5	0.4
PBE-XDM(BJ)	5.6	5.5	6.8	6.7	4.4	1.6
PBE-XDM(Z)	4.1	3.2	5.4	4.8	4.6	1.8
PBE-XDM(BJ)+ATM	2.7	0.9	3.3	2.1	3.9	-2.2
PBE-XDM(Z)+ATM	2.6	-0.1	3.2	1.0	3.9	-1.4
revPBE	19.8	-19.8	20.1	-20.1	20.3	-20.3
B86bPBE	18.6	-18.6	19.2	-19.2	19.5	-19.5
PBE	17.7	-17.7	18.3	-18.3	18.6	-18.6
PBEsol	11.9	-11.9	12.1	-12.1	12.7	-12.7
LDA	3.7	-1.9	4.0	-2.2	4.0	-2.4

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	MAE	ME	MAE	ME	MAE	ME
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revPBE-D3(0)	16.5	16.5	16.2	16.2	15.8	15.8
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revPBE-XDM(BJ)+ATM	2.2	-0.9	5.2	5.2	4.4	-2.2
revPBE-XDM(Z)+ATM	4.6	-0.3	5.3	0.8	5.3	-2.0
B86bPBE-XDM(BJ)	5.9	5.9	7.7	7.7	4.7	3.3
B86bPBE-XDM(Z)	5.8	5.5	7.3	7.1	5.3	4.5
B86bPBE-XDM(BJ)+ATM	2.3	1.2	3.5	3.2	3.1	-0.2
B86bPBE-XDM(Z)+ATM	3.0	1.3	4.0	2.6	3.5	0.4
PBE-XDM(BJ)	5.6	5.5	6.8	6.7	4.4	1.6
PBE-XDM(Z)	4.1	3.2	5.4	4.8	4.6	1.8
PBE-XDM(BJ)+ATM	2.7	0.9	3.3	2.1	3.9	-2.2
PBE-XDM(Z)+ATM	2.6	-0.1	3.2	1.0	3.9	-1.4
revPBE	19.8	-19.8	20.1	-20.1	20.3	-20.3
B86bPBE	18.6	-18.6	19.2	-19.2	19.5	-19.5
PBE	17.7	-17.7	18.3	-18.3	18.6	-18.6
PBEsol	11.9	-11.9	12.1	-12.1	12.7	-12.7
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revPBE-D3(0)	16.5	16.5	16.2	16.2	15.8	15.8
revPBE-D3(0)+ATM	10.2	10.0	9.9	9.6	9.6	9.2
PBE-D3(BJ)	14.7	14.6	14.5	14.3	12.8	12.6
PBE-D3(BJ)+ATM	8.2	7.8	8.1	7.5	6.9	6.1
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revPBE-XDM(Z)	6.5	6.1	8.2	7.8	5.4	4.2
revPBE-XDM(BJ)+ATM	2.2	-0.9	5.2	5.2	4.4	-2.2
revPBE-XDM(Z)+ATM	4.6	-0.3	5.3	0.8	5.3	-2.0
B86bPBE-XDM(BJ)	5.9	5.9	7.7	7.7	4.7	3.3
B86bPBE-XDM(Z)	5.8	5.5	7.3	7.1	5.3	4.5
B86bPBE-XDM(BJ)+ATM	2.3	1.2	3.5	3.2	3.1	-0.2
B86bPBE-XDM(Z)+ATM	3.0	1.3	4.0	2.6	3.5	0.4
PBE-XDM(BJ)	5.6	5.5	6.8	6.7	4.4	1.6
PBE-XDM(Z)	4.1	3.2	5.4	4.8	4.6	1.8
PBE-XDM(BJ)+ATM	2.7	0.9	3.3	2.1	3.9	-2.2
PBE-XDM(Z)+ATM	2.6	-0.1	3.2	1.0	3.9	-1.4
revPBE	19.8	-19.8	20.1	-20.1	20.3	-20.3
B86bPBE	18.6	-18.6	19.2	-19.2	19.5	-19.5
PBE	17.7	-17.7	18.3	-18.3	18.6	-18.6
PBEsol	11.9	-11.9	12.1	-12.1	12.7	-12.7
LDA	3.7	-1.9	4.0	-2.2	4.0	-2.4

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Functional	lightdenser		tight		PW	
	MAE	ME	MAE	ME	MAE	ME
revPBE-D3(BJ)	35.1	35.1	35.0	35.0	34.5	34.5
revPBE-D3(BJ)+ATM	27.8	27.8	27.7	27.0	27.2	27.2
revPBE-D3(0)	16.5	16.5	16.2	16.2	15.8	15.8
revPBE-D3(0)+ATM	10.2	10.0	9.9	9.6	9.6	9.2
PBE-D3(BJ)	14.7	14.6	14.5	14.3	12.8	12.6
PBE-D3(BJ)+ATM	8.2	7.8	8.1	7.5	6.9	6.1
PBE-D3(0)	9.8	9.6	9.5	9.2	8.4	8.0
PBE-D3(0)+ATM	4.1	3.2	4.0	2.8	3.7	1.8
revPBE-XDM(BJ)	5.0	5.0	11.5	11.5	6.4	4.7
revPBE-XDM(Z)	6.5	6.1	8.2	7.8	5.4	4.2
revPBE-XDM(BJ)+ATM	2.2	-0.9	5.2	5.2	4.4	-2.2
revPBE-XDM(Z)+ATM	4.6	-0.3	5.3	0.8	5.3	-2.0
B86bPBE-XDM(BJ)	5.9	5.9	7.7	7.7	4.7	3.3
B86bPBE-XDM(Z)	5.8	5.5	7.3	7.1	5.3	4.5
B86bPBE-XDM(BJ)+ATM	2.3	1.2	3.5	3.2	3.1	-0.2
B86bPBE-XDM(Z)+ATM	3.0	1.3	4.0	2.6	3.5	0.4
PBE-XDM(BJ)	5.6	5.5	6.8	6.7	4.4	1.6
PBE-XDM(Z)	4.1	3.2	5.4	4.8	4.6	1.8
PBE-XDM(BJ)+ATM	2.7	0.9	3.3	2.1	3.9	-2.2
PBE-XDM(Z)+ATM	2.6	-0.1	3.2	1.0	3.9	-1.4
revPBE	19.8	-19.8	20.1	-20.1	20.3	-20.3
B86bPBE	18.6	-18.6	19.2	-19.2	19.5	-19.5
PBE	17.7	-17.7	18.3	-18.3	18.6	-18.6
PBEsol	11.9	-11.9	12.1	-12.1	12.7	-12.7
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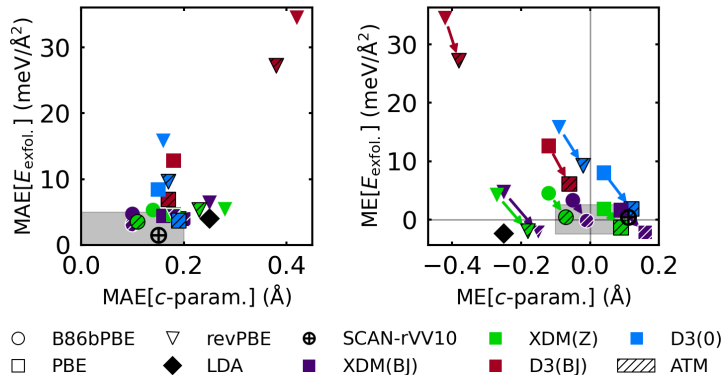
Functional	lightdenser		tight		PW	
	MAE	ME	MAE	ME	MAE	ME
revPBE-D3(BJ)	35.1	35.1	35.0	35.0	34.5	34.5
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revPBE-D3(0)	16.5	16.5	16.2	16.2	15.8	15.8
revPBE-D3(0)+ATM	10.2	10.0	9.9	9.6	9.6	9.2
PBE-D3(BJ)	14.7	14.6	14.5	14.3	12.8	12.6
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revPBE-XDM(Z)+ATM	4.6	-0.3	5.3	0.8	5.3	-2.0
B86bPBE-XDM(BJ)	5.9	5.9	7.7	7.7	4.7	3.3
B86bPBE-XDM(Z)	5.8	5.5	7.3	7.1	5.3	4.5
B86bPBE-XDM(BJ)+ATM	2.3	1.2	3.5	3.2	3.1	-0.2
B86bPBE-XDM(Z)+ATM	3.0	1.3	4.0	2.6	3.5	0.4
PBE-XDM(BJ)	5.6	5.5	6.8	6.7	4.4	1.6
PBE-XDM(Z)	4.1	3.2	5.4	4.8	4.6	1.8
PBE-XDM(BJ)+ATM	2.7	0.9	3.3	2.1	3.9	-2.2
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revPBE	19.8	-19.8	20.1	-20.1	20.3	-20.3
B86bPBE	18.6	-18.6	19.2	-19.2	19.5	-19.5
PBE	17.7	-17.7	18.3	-18.3	18.6	-18.6
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LM26: ENERGIES AND GEOMETRIES

The grey box marks acceptable error: 5 meV/Å² and 0.2 Å.

In the right (ME) plot, arrows show the shift on adding the ATM term.



- Adding ATM shifts the exfoliation energy down, and the c parameter slightly up (~ 0.06 Å)
- B86bPBE-XDM(BJ/Z)+ATM give the best simultaneous accuracy in the basis-set limit
- These approach the far more expensive non-local SCAN-rVV10 for energies, and rival or outperform it for c
- These are the lowest LM26 errors yet reported for a semi-local functional paired with an asymptotic dispersion correction

LAYERED MATERIALS: TAKEAWAYS

1. GGAs underbind, pairwise dispersion corrections overcorrect, and the large and repulsive ATM term for layered materials restores the balance
2. While BJ and Z damping are equally accurate on average, Z damping has simplicity, lesser empiricism, and a consistent united-atom limit
3. ATM has poor scaling for periodic systems and barely moves the geometry, so one could consider using it only to refine the final single-point correction rather than using during optimisation
4. B86bPBE- and PBE-XDM variants, plus SCAN-rVV10, now agree to within RPA's own error bars. Dispersion-corrected DFAs have outgrown the RPA reference data

RECAP: TOP MINIMALLY EMPIRICAL FUNCTIONALS

GMTKN55: B86bPBE0 and revPBE0

Mol. Crys.: B86bPBE0, revPBE0, and PBE0

LM26: B86bPBE and PBE

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Introduction

What's the deal with WTMAD-2?! & XDM(Z) For Molecules

Three-Body XDM(Z) for Layered Materials

Minimally Empirical Local Hybrids

- Delocalisation Error and Local Hybrid Functionals

- Construction of a New Local Hybrid Functionals: LHz & LHnz

- LHnz Performance on GMTKN54

- Ranking Against the Literature

Conclusions

DELOCALISATION ERROR

GGA functionals spread electron density too thinly. This results in:

- Too-low barrier heights, overestimated polarisabilities, spurious charge transfer
- Overbinding of halogen-bonded crystals (HalCrys4), and the SIE4x4 benchmark

These delocalisation errors can be alleviated by mixing in some amount of exact exchange, giving rise to hybrid functionals.

GLOBAL HYBRID:

$$E_{XC} = a_X E_X^{\text{HF}} + (1 - a_X) E_X^{\text{DFT}} + E_C^{\text{DFT}}$$

RANGE-SEPARATED HYBRID:

$$\frac{1}{r_{12}} = \frac{\text{erfc}(\omega r_{12})}{r_{12}} + \frac{\text{erf}(\omega r_{12})}{r_{12}}$$

THE PROBLEM WITH RANGE-SEPARATED HYBRIDS:

The optimal range-separation parameter is strongly system dependent, which limits transferability. There is growing evidence that the RS hybrids have hit their performance ceiling.

LOCAL HYBRID FUNCTIONALS

In a local hybrid, the exact-exchange fraction is position dependent:

$$E_{\text{XC}} = \int g(\mathbf{r}) \varepsilon_{\text{X}}^{\text{HF}}(\mathbf{r}) d\mathbf{r} + \int [1 - g(\mathbf{r})] \varepsilon_{\text{X}}^{\text{DFT}}(\mathbf{r}) d\mathbf{r} + E_{\text{C}}^{\text{DFT}},$$

so we can use more exact exchange in areas of large electron delocalization. However, many top local hybrids tend to be highly empirical:

Functional	Type	N_{param}
ω LH25tdE-D4	RS + local	22
LH24n-D4	local	neural network
B22plus	local	20

THE CHALLENGE:

How accurate can we make a local hybrid by focusing on sound physics, with:

No range separation

No calibration functions

No B97-like power series

No neural networks

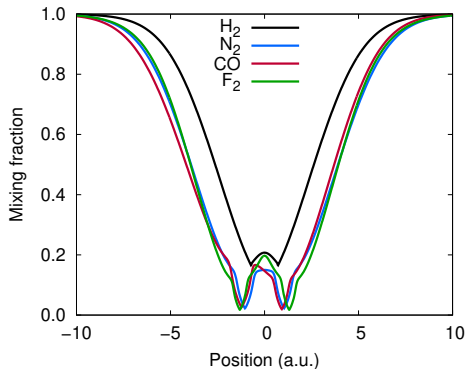
CONSTRUCTION OF A NEW LOCAL HYBRID FUNCTIONALS: LHZ & LHNZ

Based off B86bPBE0-XDM(Z), a top and transferable global hybrid with solid physics, despite being minimally empirical.

$$E_{XC} = E_X^{B86b} + \sum_{\sigma} \int g_{\sigma}(\mathbf{r}) \left[\varepsilon_{X,\sigma}^{\text{exact}}(\mathbf{r}) - \varepsilon_{X,\sigma}^{B86b}(\mathbf{r}) \right] d\mathbf{r} \\ + E_C^{\text{PBE}} + E_{\text{disp}}^{\text{XDM(Z)}}$$

For B86bPBE0, $g_{\sigma}(\mathbf{r}) = 0.25$ everywhere. For our local hybrid, we base exchange-energy mixing on the σ -spin correlation length, $z_{\sigma}(\mathbf{r})$, which avoids rapid oscillations near the nuclei unlike ratios of kinetic-energy densities. For LHZ:

$$g_{\sigma}(\mathbf{r}) = \text{erf}\{c z_{\sigma}(\mathbf{r})\} \quad z_{\sigma}(\mathbf{r}) = \left| \frac{\rho_{\sigma}(\mathbf{r})}{\varepsilon_{X,\sigma}^{B86b}(\mathbf{r})} \right|$$



Correlation length and mixing function ($c = 0.10$) for chosen diatomic molecules along their bond axis.

E. R. Johnson, *J. Chem. Phys.* **141**, 124120. (2014) doi: 10.1063/1.4896302

E. R. Johnson & K. R. Bryenton, Submitted to: *J. Chem. Phys.* (2026). doi: 10.48550/arXiv.2609.06162

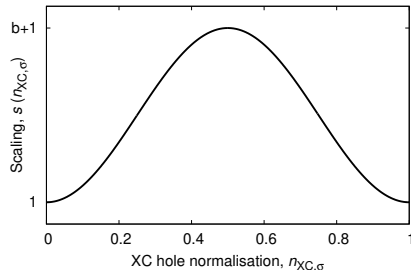
CAPTURING FRACTIONAL CHARGE FOR LHNZ

The correlation length does not predict the presence of delocalization error alone. However, XC hole normalisation, $n_{XC,\sigma}$ from Becke's 2003 model, does. Thus, let's introduce a scaling function for LHNz:

$$g_{\sigma}(\mathbf{r}) = \text{erf} \left\{ c \left[\underbrace{b \sin^2(\pi n_{XC,\sigma}) + 1}_{s(n_{XC,\sigma})} \right] z_{\sigma}(\mathbf{r}) \right\}$$

$s = 1$ when $n_{XC,\sigma}$ is an integer, reducing LHNz to LHz

$s = b + 1$ when $n_{XC,\sigma} = 0.5$ is the maximum enhancement



ESCAPING THE ZERO-SUM GAME:

Usually, fixing fractional-charge error costs fractional-spin error. But $s = 1$ for the neutral, spin-averaged H atom, so LHNz gives the same H_2 dissociation limit as LHz. Strong-correlation is needed to recover the exact spin-unpolarised limit.

Mol.	Atom	$n_{XC,\alpha}$	$n_{XC,\beta}$	s_{α}	s_{β}
<i>Integer Charge</i>					
H_2	H	1	0	1	1
H_2	H	1	1	1	1
He_2	He	1	1	1	1
<i>Fractional Charge</i>					
H_2^+	$H^{+0.5}$	0.5	0	$b + 1$	1
He_2^+	$He^{+0.5}$	1	0.5	1	$b + 1$

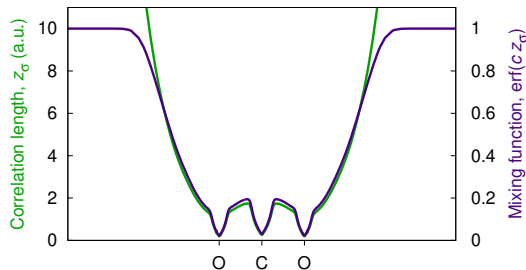
LHNZ'S "FINE-TUNING" PARAMETERS

$$g_{\sigma}(\mathbf{r}) = \text{erf} \left\{ c \left[b \sin^2(\pi n_{\text{XC},\sigma}) + 1 \right] z_{\sigma}(\mathbf{r}) \right\}$$

The complete parameter list for LHNz:

	Parameter	Value
1.	c	0.10
2.	b	4.6
3.	z_{damp}	117237 Ha ⁻¹

Using PBE0/def2-QZVP densities in postG

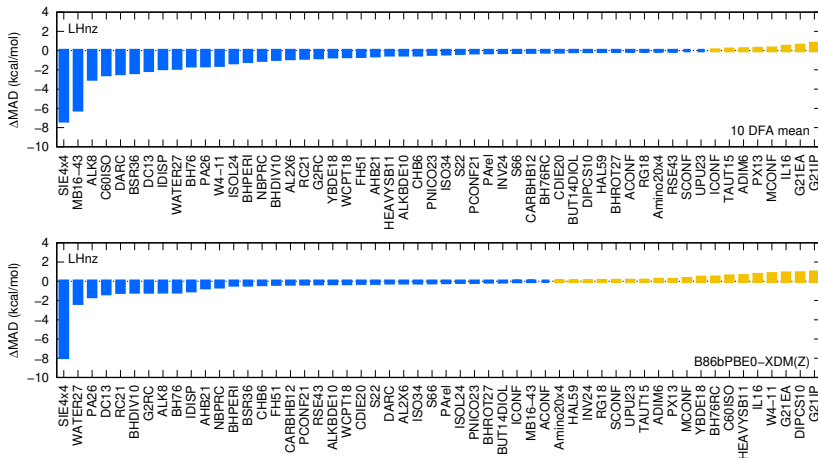


Correlation length and mixing function ($c = 0.10$) along the CO_2 bond axis

In keeping with Axel's philosophy, the c and b terms are "fine tuning" parameters:

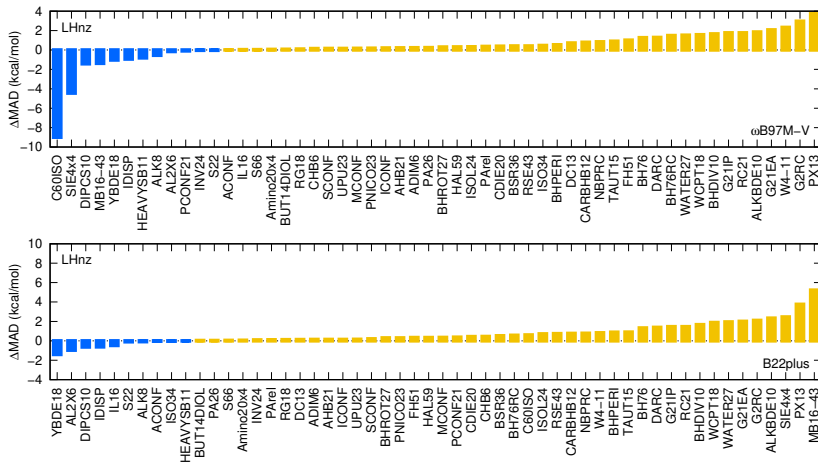
- $c \ll 1/2$: large exact-exchange fractions are inappropriate in bonding regions, where $z_{\sigma} \approx 2$ for CO_2
- $b \sim 4$: the exact-exchange fraction should go from $\sim 25\% \rightarrow 100\%$ in regions of fractional charge

LHnz vs THE CONSENSUS AVERAGE AND ITS PARENT FUNCTIONAL



- **vs 10-DFA:** consistently good, with standout results for MB16-43 and especially SIE4x4, where the MAD more than halves from 13.6 to 6.31 kcal/mol
- **vs B86bPBE0-XDM(Z):** gains sit exactly where delocalisation error lives: SIE4x4, WATER27, PA26, DC13, RC21, BHDIV10, but slightly worse performance for ionic systems: G21IP, G21EA, DIPCS10, and IL16

LHnz VS THE BEST IN THE LITERATURE



- **vs ωB97M-V:** (12 params): ωB97M-V is somewhat more accurate for most benchmarks, especially G21EA, W4-11, G2RC, and PX13, but LHnz is decisively better on C60ISO and SIE4x4
- **vs B22plus:** (20 params): LHnz can't compete here without the explicit non-dynamical correlation contributions that B22plus carries

TOP 32/114 HYBRID FUNCTIONALS ON GMTKN54

Rank	Functional	Type	N_{param}	WTMAD- N		$N_{r>2}$	Ratio outliers			Difference outliers		
				2	4		MAX	r	Set	$N_{d>5}$	MAX	d Set
1	ω LH25tdE-D4	RS+local	22	2.57	3.32	1	2.55	IL16		1	3.69	DIPCS10
2	LH24n-D4	local	NN	2.91	3.54	1	2.17	IL16		1	2.27	DIPCS10
3	B22plus	local	20	2.76	3.63	1	2.64	IL16		0	0.91	IL16
4	ω B97M-V	RS	12	3.08	3.74	1	2.22	C60ISO		1	6.51	C60ISO
5	LH25nP-D4	local	NN	2.44	3.79	3	3.03	DIPCS10		2	9.67	C60ISO
6	ω LH23tdE-D4	RS+local	6	3.58	4.07	1	3.12	IL16		0	1.18	DIPCS10
7	B22	local	9	3.31	4.21	0	1.91	ADIM6		0	1.41	DIPCS10
8	ω B97M-D4	RS	15	3.69	4.30	1	2.24	C60ISO		2	6.60	C60ISO
9	ω B97X-V	RS	10	3.84	4.44	1	2.57	C60ISO		2	13.21	MB16-43
10	ω B97M-D3(BJ)	RS	14	3.77	4.61	3	2.47	C60ISO		2	7.86	C60ISO
11	LH20t-D4	local	12	4.46	4.62	1	2.29	IL16		1	2.79	DIPCS10
12	ω B97X-D4	RS	13	3.96	4.63	1	2.65	C60ISO		1	8.83	C60ISO
13	B22-10b	local	7	3.60	4.68	1	3.12	ADIM6		0	1.88	HEAVYSB11
14	scLH22t-D4	local	14	4.37	4.71	1	2.58	IL16		2	3.07	DIPCS10
15	ω B97M-D3(0)	RS	14	4.10	4.73	1	2.31	C60ISO		2	7.01	C60ISO
16	B22-10a	local	7	3.70	4.83	2	2.31	ADIM6		2	2.55	DIPCS10
17	LH24x-D4	local	11	4.83	5.09	3	3.07	IL16		0	1.16	DIPCS10
18	LH24x-core-D4	local	9	4.91	5.18	3	3.24	ADIM6		0	1.45	DIPCS10
19	LHnz	local	3	4.90	5.19	1	2.03	ADIM6		0	0.73	G21IP
20	PW6B95-D3(BJ)	global	8	5.45	5.19	1	2.31	IL16		0	1.76	SIE4x4
21	M05-2X-D3(0)	global	25	4.44	5.21	1	2.08	ADIM6		3	7.02	MB16-43
22	M06-2X-D3(0)	global	35	4.72	5.22	2	3.50	HEAVYSB11		1	5.83	HEAVYSB11
23	LH24x-SC-D4	local	9	5.05	5.22	3	3.76	ADIM6		0	1.82	SIE4x4
24	ω B97X-D3(BJ)	RS	12	4.07	5.26	2	2.99	C60ISO		3	10.65	C60ISO
25	PW6B95-V	global	6	5.50	5.28	2	2.49	ADIM6		0	1.81	SIE4x4
26	LHz	local	2	5.10	5.30	1	2.03	ADIM6		0	0.64	G21IP
27	PW6B95-D3(0)	global	9	5.50	5.33	1	3.18	ADIM6		0	1.83	SIE4x4
28	revPBE0-D3(BJ)	global	2	5.21	5.34	0	1.51	W4-11		1	3.03	W4-11
29	GH-PBE _x -B95c-D4	global	8	5.24	5.35	1	2.20	ADIM6		0	1.11	DIPCS10
30	revPBE0-XDM(Z)	global	1	5.29	5.38	0	1.48	W4-11		1	2.83	W4-11
31	ω B97X-D3(0)	RS	12	4.65	5.43	1	2.54	C60ISO		2	17.20	MB16-43
32	B86bPBE0-XDM(Z)	global	1	5.55	5.44	0	1.18	RSE43		0	0.61	SIE4x4

Introduction

What's the deal with WTMAD-2?! & XDM(Z) For Molecules

Three-Body XDM(Z) for Layered Materials

Minimally Empirical Local Hybrids

Conclusions

Concluding Remarks

Future Work

Acknowledgements

CONCLUDING REMARKS

1. WTMAD-2 is a misleading summary of a functional's performance, with the top 3 benchmarks weighing as much as the bottom 36, and WTMAD-4 is a much fairer representation
2. XDM(Z) needs only one parameter and is accurate, consistent, and transferable across GMTKN55, molecular crystals, and layered materials, especially when paired with B86bPBE(0)
3. Three-body interactions matter for layered materials. XDM+ATM, with either damping, gives the lowest LM26 errors yet reported for a semi-local functional with an asymptotic dispersion correction
4. Better reference data, perhaps QMC, is required for the continued study of layered materials. Dispersion-corrected GGA's are now inside RPA's own error bars
5. LHNz beats every global hybrid in the literature with only three parameters with no large outliers

MAIN TAKEAWAY:

Minimally-empirical functionals with good physics are still able to perform accurately, consistently, and with fewer outliers than the heavily parametrised alternatives.

FUTURE WORK

1. **Better correlation models for LHnz:** Full long-range exact exchange with no compensating non-local correlation gives a systematic error for ionic systems
2. **Non-dynamical correlation terms:** While strong correlation is necessary to recover the exact spin-unpolarised H_2 dissociation limit, non-dynamical correlation terms will help and should improve performance for MB16-43 and a number of other benchmarks where B22plus excels
3. **Wider adoption of exchange-energy density in DFT codes:** The exchange-energy density is still unavailable in most production codes, which is the main barrier to local hybrid functionals gaining usage

XDM(Z) AVAILABILITY:

XDM with Z damping is implemented and freely available in:
FHI-aims, Quantum ESPRESSO, and postg.

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- Kanav Sharma
- Miko Crouse



QUESTIONS?

WANT MY SLIDES?



[HTTPS://GITHUB.COM/KYLEBRYENTON/SLIDES-POSTERS](https://github.com/KyleBryenton/slides-posters)

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THE XDM MODEL

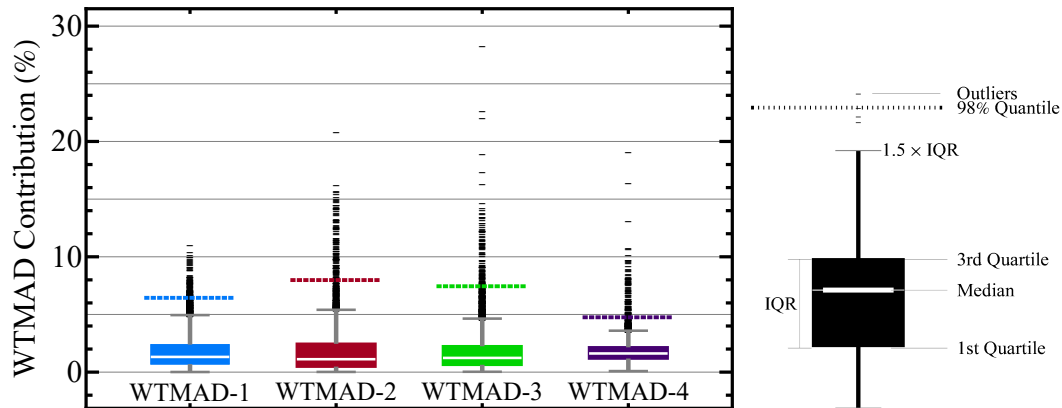
These exchange-hole dipole moments are used along with Hirshfeld weights, w_i , and the electron density, ρ_σ , to compute the multipole moment integrals,

$$\langle M_\ell^2 \rangle = \sum_\sigma \int w_i(\mathbf{r}) \rho_\sigma(\mathbf{r}) \left[r^\ell - (r - d_{X\sigma})^\ell \right]^2 d\mathbf{r}.$$

These are then combined with atom-in-molecule polarisabilities, α_i , to solve for the heteroatomic dispersion coefficients,

$$\begin{aligned} C_{6,ij}^{\text{XDM}} &= \frac{\alpha_i \alpha_j \langle M_1^2 \rangle_i \langle M_1^2 \rangle_j}{\alpha_i \langle M_1^2 \rangle_j + \alpha_j \langle M_1^2 \rangle_i} \\ C_{8,ij}^{\text{XDM}} &= \frac{3}{2} \frac{\alpha_i \alpha_j \left(\langle M_1^2 \rangle_i \langle M_2^2 \rangle_j + \langle M_2^2 \rangle_i \langle M_1^2 \rangle_j \right)}{\alpha_i \langle M_1^2 \rangle_j + \alpha_j \langle M_1^2 \rangle_i} \\ C_{10,ij}^{\text{XDM}} &= 2 \frac{\alpha_i \alpha_j \left(\langle M_1^2 \rangle_i \langle M_3^2 \rangle_j + \langle M_3^2 \rangle_i \langle M_1^2 \rangle_j \right)}{\alpha_i \langle M_1^2 \rangle_j + \alpha_j \langle M_1^2 \rangle_i} + \frac{21}{5} \frac{\alpha_i \alpha_j \langle M_2^2 \rangle_i \langle M_2^2 \rangle_j}{\alpha_i \langle M_1^2 \rangle_j + \alpha_j \langle M_1^2 \rangle_i}. \end{aligned}$$

WTMAD BoxPLOTS



THE 2- AND 3-BODY XDM DISPERSION ENERGY

When applied to periodic systems, the pairwise XDM energy expression becomes

$$E_{\text{XDM}}^{(2)} = -\frac{1}{2} \sum_{\mathbf{L}} \sum_{i \neq j'} \sum_{n=6,8,10} \frac{C_{n,ij} f_n^{\text{BJ/Z}}(R_{ij})}{R_{ij,\mathbf{L}}^n},$$

and, similarly for the ATM contribution,

$$E_{\text{XDM}}^{(3)} = \frac{1}{6} \sum_{\mathbf{L}, \mathbf{M}} \sum_{\substack{i \neq j' \neq k' \\ j \neq k''}} \frac{C_{9,ijk} (3 \cos \theta_i \cos \theta_j \cos \theta_k + 1)}{(R_{ij,\mathbf{L}}^3)(R_{ik,\mathbf{M}}^3)(R_{jk,\mathbf{M}-\mathbf{L}}^3)} \times f_{\text{ATM}}^{\text{BJ/Z}}(R_{ij,\mathbf{L}}, R_{ik,\mathbf{M}}, R_{jk,\mathbf{M}-\mathbf{L}}).$$

$\mathbf{L}$ and $\mathbf{M}$ are displacement lattice vectors that allow atoms j and k to be in different cells. The prime excludes the $i = j$ and $i = k$ term for lattice vector $\mathbf{L} = 0$ and the double prime excludes the $j = k$ term for all lattice vectors $\mathbf{L} = \mathbf{M}$. Thus, the total energy is

$$E_{\text{XDM}}^{\text{tot}} = E_{\text{XDM}}^{(2)} + E_{\text{XDM}}^{(3)}.$$

